

Cardiovascular Risk Prediction Using Machine Learning

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Abstract—The prediction of cardiovascular risk will help tell which individuals are prone to development of serious cardiovascular complications. The goal of this project is to explain how machine learning can be used for cardiovascular risk prediction. The dataset used consists of 1,529 individual records and 22 attributes. The cardiovascular risk is classified into three categories: LOW, INTERMEDIARY, and HIGH. Data preprocessing procedure includes working with missing values, passing through numerical and categorical data, retrieving the clinically important information, and eliminating variables that can cause target leakage. Logistic Regression, Random Forest, AdaBoost, and Neural Networks are used as baseline models and compared with KNN and Gradient Boosting after dimension reduction by means of PCA. Additionally, a parallel regression model predicting the actual continuous CVD Risk Score is created. Based on the experiments, we can see that Logistic Regression outperforms the other classifiers in terms of its test accuracy which is about 65.36%. In second place remains Gradient Boosting (65.03%) and AdaBoost in third (64.38%). The class-wise performance says that all the classifiers have problems determining the LOW-risk class that is significantly smaller than the rest categories, while the regression approach achieved R^2 of 0.951 for the same target, which suggests that accuracy alone is not enough to assess the algorithms. Thus the project aims to compare multiple machine-learning methods and their class-specific performance.

Index Terms—Cardiovascular Risk Prediction, Machine Learning, Logistic Regression, Random Forest, AdaBoost, Neural Network, Principal Component Analysis, Multiclass Classification

I. INTRODUCTION

Cardiovascular diseases (CVDs) are disorders affecting the heart and blood vessels and remain an important healthcare concern worldwide. Many cardiovascular complications are associated with measurable risk factors such as age, blood pressure, cholesterol, diabetes, smoking, body mass index (BMI), physical activity, and family history. Identifying individuals at different levels of cardiovascular risk can therefore support earlier preventive measures and better clinical decision-making. Traditional risk assessment methods such as QRISK2 and the American Heart Association PREVENT equations estimate cardiovascular risk using combinations of established clinical factors [1], [2].

Machine learning provides another approach to cardiovascular risk assessment by learning patterns and relationships

directly from patient data. Unlike a single predefined statistical equation, different machine learning algorithms can capture different relationships among demographic, clinical, and lifestyle variables. Previous studies have applied algorithms such as Logistic Regression, Random Forest, Support Vector Machine, AdaBoost, Neural Networks, XGBoost, and ensemble methods for cardiovascular risk prediction [3]–[5]. However, their performance varies depending on the population, dataset size, selected features, class distribution, and prediction objective.

Recent studies have also examined cardiovascular risk prediction in Bangladeshi populations. Islam et al. compared several machine learning algorithms using BDHS 2022 data and found XGBoost to provide the strongest performance for identifying individuals at high cardiovascular risk [10]. Tasnim et al. used clinical data from 1,529 Bangladeshi patients and applied a stacking regression framework with explainable AI for predicting cardiovascular risk levels [12]. These studies indicate the potential of machine learning for cardiovascular risk assessment in a local context while also showing the need to compare different modeling strategies.

This project investigates cardiovascular risk prediction as a multiclass classification problem with three categories: LOW, INTERMEDIARY, and HIGH risk. The available dataset contains structured clinical, demographic, and lifestyle information from 1,529 individuals. Logistic Regression, Random Forest, AdaBoost, and a Neural Network are initially implemented and compared using overall and class-wise evaluation measures. The objective is to determine how effectively these machine learning models can distinguish different cardiovascular risk levels and to identify the limitations that affect prediction across the three classes.

II. LITERATURE REVIEW

Cardiovascular risk prediction has been investigated using traditional statistical equations, conventional machine learning algorithms, deep learning, and ensemble-based approaches. Previous research provides useful guidance regarding relevant cardiovascular predictors, model selection, multiclass risk stratification, evaluation procedures, and limitations that

should be considered when developing machine learning models for cardiovascular risk assessment.

Hippisley-Cox et al. developed QRISK2 using Cox proportional hazards regression on approximately 2.3 million patients and reported ROC statistics of approximately 0.817 for women and 0.792 for men [1]. The model is relevant to our project because several of its predictors, including age, smoking, blood pressure, cholesterol, BMI, and diabetes, correspond to common cardiovascular risk attributes. However, QRISK2 predicts future events using a longitudinal survival model rather than the LOW, INTERMEDIARY, and HIGH multiclass risk formulation used in our project.

Khan et al. developed the American Heart Association PREVENT equations using data from more than six million adults and reported external-validation C-statistics of approximately 0.794 for women and 0.757 for men for total CVD [2]. The study supports the importance of routinely available cardiovascular, metabolic, and kidney-related predictors and provides a strong conventional risk-prediction reference for our work. However, PREVENT is a survival-based risk equation designed for a large US population rather than a machine learning multiclass classifier.

Unnikrishnan et al. compared the Framingham approach, locally trained linear models, and Support Vector Machine for cardiovascular risk assessment and found that SVM provided improved sensitivity and specificity [3]. This is relevant to our project because the study uses common clinical factors such as cholesterol, blood pressure, BMI, smoking, diabetes, and hypertension for machine learning-based risk prediction. The study used a specific local cohort and did not evaluate a broader multiclass cardiovascular risk framework.

Cho et al. compared Logistic Regression, TreeBag, Random Forest, AdaBoost, and Neural Networks using 222,998 Korean adults, with the Neural Network achieving the highest C-statistic of 0.751 [4]. This study is especially relevant because several of the same algorithms are currently implemented in our project and it directly compares machine learning models for cardiovascular risk prediction. Its population consisted specifically of statin-naïve Korean adults, so its reported performance may not generalize directly to other populations or datasets.

Alaa et al. applied AutoPrognosis to 423,604 UK Biobank participants and achieved an AUC-ROC of 0.774, compared with 0.724 for the Framingham score [5]. The study demonstrates that machine learning can combine traditional and non-traditional predictors to improve cardiovascular risk prediction and supports systematic model comparison in our project. However, the framework used hundreds of candidate variables and a very large cohort, making it substantially more complex than our smaller structured dataset.

Li et al. evaluated 19 statistical and machine learning approaches using approximately 3.6 million patients and found similar population-level C-statistics of around 0.87 despite considerable differences in individual risk predictions [6]. This finding is relevant because it demonstrates that model selection should consider prediction consistency and risk estimation

rather than relying only on a single overall performance measure. The work primarily addressed long-term event prediction with censoring, whereas our current project evaluates categorical cardiovascular risk levels.

Barbieri et al. combined deep learning and survival analysis to estimate five-year cardiovascular risk among more than 2.1 million New Zealand adults and found that the deep learning models outperformed conventional Cox models in explained variance, calibration, and discrimination [7]. The study supports investigating nonlinear relationships among cardiovascular predictors and demonstrates the potential advantage of more flexible learning algorithms. However, the administrative dataset did not contain several direct clinical measurements such as blood pressure and lipid results that are available in our structured dataset.

Neumann et al. developed a cardiovascular risk model for healthy adults aged 70 years or older using Cox regression with LASSO-supported variable selection and obtained an AUC of 0.681 in development and 0.642 during external validation [8]. The study is relevant because it identifies age, smoking, systolic blood pressure, cholesterol, creatinine, and diabetes as useful cardiovascular predictors and demonstrates the importance of external validation. Its restriction to older individuals limits direct generalization to datasets containing a broader age distribution.

Suri et al. reviewed 265 studies involving multiclass, multi-label, and ensemble machine learning approaches for cardiovascular risk stratification and found substantial use of feature selection and ensemble-based methods [9]. This review is highly relevant because our target contains three cardiovascular risk categories and therefore fits directly within the multiclass risk-stratification paradigm discussed by the authors. Since it is a narrative review rather than a newly developed predictive model, it does not provide a single dataset or experimental result that can be directly reproduced.

Islam et al. used BDHS 2022 data from 2,221 Bangladeshi individuals and compared Logistic Regression, Naïve Bayes, ANN, Random Forest, and XGBoost, with XGBoost achieving 68.22% accuracy and an AUC of 0.721 [10]. The study is strongly relevant because it focuses specifically on cardiovascular risk prediction in Bangladesh and also applies feature selection and imbalance handling before model comparison. However, its high-risk outcome was constructed mainly from hypertension and diabetes information and the cross-sectional dataset limits longitudinal risk interpretation.

Shah et al. proposed a hybrid ensemble incorporating models such as Gradient Boosting, CatBoost, LightGBM, SVM, Neural Networks, and an XGBoost meta-learner, achieving an AUC-ROC of 0.82 [11]. The study is relevant because it demonstrates how ensemble learning and SHAP-based explainability can improve both cardiovascular prediction and interpretation of influential risk factors. Its multi-model architecture and combination of multiple datasets introduce considerably greater complexity than the baseline models currently used in our project.

Tasnim et al. developed a stacking regression framework

TABLE I
CVD RISK LEVEL CLASS DISTRIBUTION

Risk Level	Count	Percentage
HIGH	728	47.6%
INTERMEDIARY	581	38.0%
LOW	220	14.4%
Total	1,529	100%

combining Ridge Regression, Theil-Sen Regression, and Gradient Boosting on a dataset of 1,529 Bangladeshi patients and obtained an R^2 of approximately 0.96 under cross-validation [12]. This study is particularly relevant because its dataset structure and LOW, INTERMEDIARY, and HIGH cardiovascular risk formulation closely match our project and it also identifies BMI, diabetes, and blood pressure as influential factors through SHAP. However, the categorical risk levels were treated through a regression-based framework and the data originated from a single institution, leaving scope to investigate the same risk problem explicitly through multiclass classification and class-wise evaluation.

III. DATASET AND PREPROCESSING

A. Dataset Description

The dataset consists of structured clinical, demographic, and lifestyle records for 1,529 individuals described by 22 attributes, including physical measurements (age, weight, height, BMI, abdominal circumference), clinical measurements (blood pressure, cholesterol, HDL, fasting blood sugar, estimated LDL), and lifestyle/history variables (smoking status, diabetes status, physical activity, family history of CVD). Each record carries two possible targets: a categorical CVD Risk Level (LOW, INTERMEDIARY, HIGH) used for the main classification task, and a continuous CVD Risk Score used for a supplementary regression task. Table I shows a clear class imbalance, with the LOW-risk group making up only 14.4% of records.

B. Data Cleaning, Feature Recovery, and Preprocessing

Initial inspection showed no duplicate records but moderate missingness across several columns (e.g., 82 missing Diastolic BP, 81 Weight, 80 HDL). Recoverable values were reconstructed rather than discarded: blood pressure was parsed from a combined string field, missing height in meters was derived from height in centimeters, and BMI was recalculated from weight and height where both were available; physiologically invalid negative Estimated LDL values were treated as missing. The combined blood-pressure string, the duplicate height column, and both risk targets were then dropped to prevent leakage, leaving 18 candidate features (12 numerical, 6 categorical).

Exploratory analysis showed most numerical features close to uniformly distributed, with BMI and the CVD Risk Score closer to bell-shaped; Sex was nearly balanced (773 female, 756 male). A correlation analysis confirmed strong redundancy between height in meters and centimeters ($r = 1.00$), Total

Cholesterol and Estimated LDL ($r = 0.96$), and Abdominal Circumference and Waist-to-Height Ratio ($r = 0.90$), while BMI, Total Cholesterol, Systolic BP, and Estimated LDL showed the strongest linear relationships with the CVD Risk Score.

The cleaned feature set was split 80/20 (stratified, `random_state = 42`) into 1,223 training and 306 testing records. A scikit-learn `ColumnTransformer` pipeline, fit only on the training data, median-imputed and standardized numerical features and most-frequent-imputed and one-hot-encoded categorical features, producing a 27-dimensional processed feature space.

IV. METHODOLOGY

A. Logistic Regression

A multinomial Logistic Regression classifier (scikit-learn, `max_iter = 1000`) was fit on the preprocessed features as an interpretable linear baseline.

B. Random Forest (Implemented from Scratch)

A Random Forest was implemented from scratch on top of `DecisionTreeClassifier` base learners to control the bagging and feature-randomization logic explicitly. Each of 25 trees (max depth 5) is fit on a bootstrap sample using a random subset of $\sqrt{n}$ features per split; predictions are decided by majority vote.

C. AdaBoost (Implemented from Scratch)

A multiclass AdaBoost classifier (SAMME algorithm) was implemented from scratch using decision stumps (`max_depth = 1`) as weak learners. Sample weights are reweighted after each of 50 boosting rounds based on the stump's weighted error, and each stump's amount-of-say is

$$\alpha = \ln \left(\frac{1 - \text{err}}{\text{err}} \right) + \ln(K - 1), \quad (1)$$

for $K = 3$ classes; stumps at or above chance level are rejected. The final prediction uses the highest total α -weighted vote.

D. Neural Network

A feed-forward network (Keras/TensorFlow) with one hidden Dense layer of 32 ReLU units and a 3-unit softmax output was trained with the Adam optimizer ($lr = 0.001$) and sparse categorical cross-entropy loss for 100 epochs (batch size 32), using an internal 70/15 training/validation split alongside the untouched 306-record test set.

E. Dimensionality Reduction with PCA

PCA was fit on the training data and configured to retain components explaining at least 95% of variance, retaining 16 of 27 components (96.04% of variance). All four baseline models were retrained on this 16-dimensional representation.

TABLE II
HYPERPARAMETER CONFIGURATION FOR EACH MODEL

Model	Key Hyperparameters
Logistic Regression	max_iter = 1000
Random Forest (scratch)	n_estimators=25, max_depth=5, max_features= $\sqrt{n}$
AdaBoost (scratch)	n_estimators=50, stump_depth=1, SAMME
Neural Network	32 ReLU units, Adam ($lr = 0.001$), 100 epochs, batch=32
KNN	$k = 15$, distance-weighted
Gradient Boosting	n_estimators=100, lr=0.1, max_depth=3
RF Regressor	n_estimators=200, max_depth=8

TABLE III
OVERALL TEST ACCURACY OF THE FOUR BASELINE MODELS

Model	Test Accuracy
Logistic Regression	65.36%
Random Forest	58.82%
AdaBoost	64.38%
Neural Network	62.42%

F. Additional Models on PCA-Reduced Features

Two further scikit-learn classifiers were trained on the PCA-reduced data: K-Nearest Neighbors ($k = 15$, distance-weighted) and Gradient Boosting (100 estimators, learning rate 0.1, max depth 3).

G. Regression Model for CVD Risk Score

A Random Forest Regressor (200 estimators, max depth 8) was trained within the same pipeline to predict the continuous CVD Risk Score directly, using the records with a valid score (80/20 split: 1,167 training, 292 testing), isolating the difficulty introduced by discretizing the score into three risk levels.

V. EXPERIMENTAL SETUP

A. Training Configuration

All classifiers used the same 80/20 stratified split (random_state = 42); the Neural Network additionally used its internal 70/15 split for monitoring convergence. Table II summarizes the key hyperparameters used for each model.

B. Evaluation Metrics

Classification performance is reported using overall test accuracy plus per-class precision, recall, and F1-score, since Table I's class imbalance makes accuracy alone potentially misleading; confusion matrices were also inspected. Regression performance is reported using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

VI. RESULTS AND ANALYSIS

A. Overall Performance Comparison

Table III presents the test accuracy of the four baseline models. Logistic Regression achieves the highest accuracy (65.36%), followed by AdaBoost (64.38%), the Neural Network (62.42%), and Random Forest (58.82%).

TABLE IV
CLASS-WISE PRECISION (P) AND RECALL (R)

Model	HIGH		INTERMEDIARY		LOW	
	P	R	P	R	P	R
LR	0.71	0.75	0.63	0.74	0.31	0.09
RF	0.61	0.74	0.56	0.62	0.00	0.00
AdaBoost	0.66	0.77	0.63	0.73	0.00	0.00
NN	0.70	0.72	0.61	0.66	0.30	0.20

TABLE V
EFFECT OF PCA ON TEST ACCURACY

Model	Before PCA	After PCA	Δ (pts)
Logistic Regression	65.36%	64.71%	−0.65
Random Forest	58.82%	59.48%	+0.65
AdaBoost	64.38%	60.78%	−3.59
Neural Network	62.42%	61.76%	−0.65

B. Class-wise Performance Analysis

Table IV breaks down precision and recall by class. All four models identify HIGH and INTERMEDIARY reasonably well (recall 0.62–0.77), but the LOW class is identified poorly by every model: Random Forest and AdaBoost achieve 0.00 recall for LOW, while Logistic Regression recalls only 9%. The Neural Network is a partial exception (0.30 precision, 0.20 recall on LOW), despite its lower overall accuracy—illustrating that accuracy alone understates its value for the minority class.

The confusion matrices in Fig. 1 show that Random Forest and AdaBoost never predict LOW at all, sending every LOW-risk patient to HIGH or INTERMEDIARY, and that most LOW-class errors across all models are confused with INTERMEDIARY rather than HIGH—the main source of error is the LOW/INTERMEDIARY boundary. The Neural Network showed a widening train-validation accuracy gap over 100 epochs (training accuracy ≈ 0.80 , validation plateauing near 0.62–0.66), consistent with its reported test accuracy of 62.42%.

C. Effect of Dimensionality Reduction (PCA)

Table V compares baseline accuracy before and after re-training on the 16-component PCA representation (96.04% of variance retained). The effect is small and inconsistent: Random Forest improves marginally (+0.65 pts), Logistic Regression and the Neural Network drop slightly (−0.65 pts each), while AdaBoost loses more (−3.59 pts)—suggesting most of the 27 processed dimensions carry non-redundant information despite the strong pairwise correlations noted in Section III-B, so compression provides little benefit and can mildly hurt feature-sensitive models such as AdaBoost.

Table VI ranks all six evaluated classifiers by test accuracy, adding the two PCA-based models from Section IV-F. Gradient Boosting on PCA-reduced features reaches 65.03%, essentially matching Logistic Regression and outperforming every other model except it; KNN performs worse (59.80%), close to

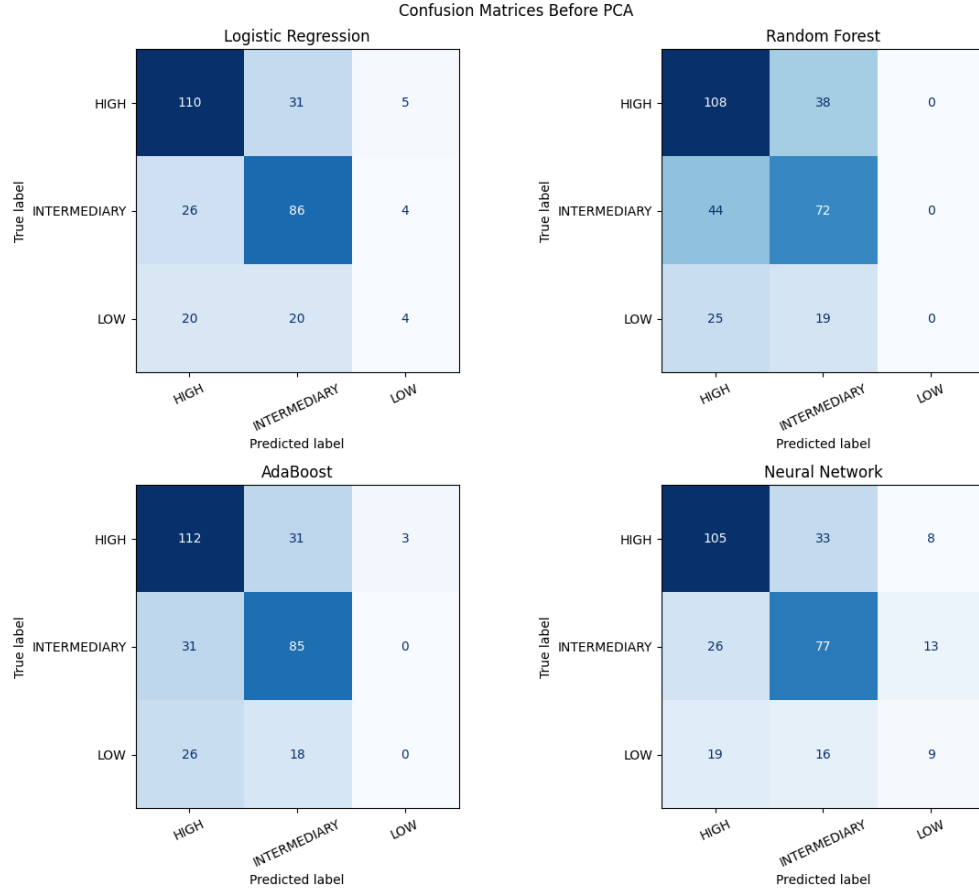


Fig. 1. Confusion matrices of the four baseline models (before PCA) on the test set.

TABLE VI
OVERALL COMPARISON ACROSS ALL SIX EVALUATED MODELS

Model	Test Accuracy
Logistic Regression	65.36%
Gradient Boosting (PCA)	65.03%
AdaBoost	64.38%
Neural Network	62.42%
KNN (PCA)	59.80%
Random Forest	58.82%

TABLE VII
MACRO-AVERAGE ROC AUC BY MODEL

Model	Macro-Avg AUC
Neural Network — After PCA	0.77
Gradient Boosting — After PCA	0.76
Neural Network — Before PCA	0.75
Logistic Regression — Before PCA	0.73
Logistic Regression — After PCA	0.73
KNN — After PCA	0.72

Random Forest. After PCA, Random Forest still never predicts LOW, AdaBoost’s LOW recall improves only marginally (4 of 44 correct), and the Neural Network and Gradient Boosting produce the most balanced confusion matrices, though INTERMEDIARY remains the dominant confusion for LOW across every classifier.

D. ROC Curve Analysis

To evaluate the models beyond a fixed 0.5 threshold, one-vs-rest ROC curves and per-class AUC were computed for Logistic Regression and the Neural Network (before and after PCA) together with KNN and Gradient Boosting after PCA. Fig. 2 shows the resulting curves, and Table VII ranks the six model/PCA combinations by macro-average AUC. The

Neural Network after PCA achieves the highest macro AUC (0.77), followed by Gradient Boosting (0.76) and the Neural Network before PCA (0.75); both Logistic Regression variants and KNN cluster around 0.72–0.73.

Notably, the LOW class has the lowest per-class AUC for the linear models (0.61 for Logistic Regression) but a substantially higher AUC for the Neural Network after PCA (0.75) and Gradient Boosting (0.71). This gap between AUC and the near-zero recall values in Table IV indicates several models—especially Logistic Regression—do rank LOW-risk patients reasonably well by predicted probability, but the default 0.5 threshold is too high for the minority class to ever be selected as the arg-max prediction, pointing toward

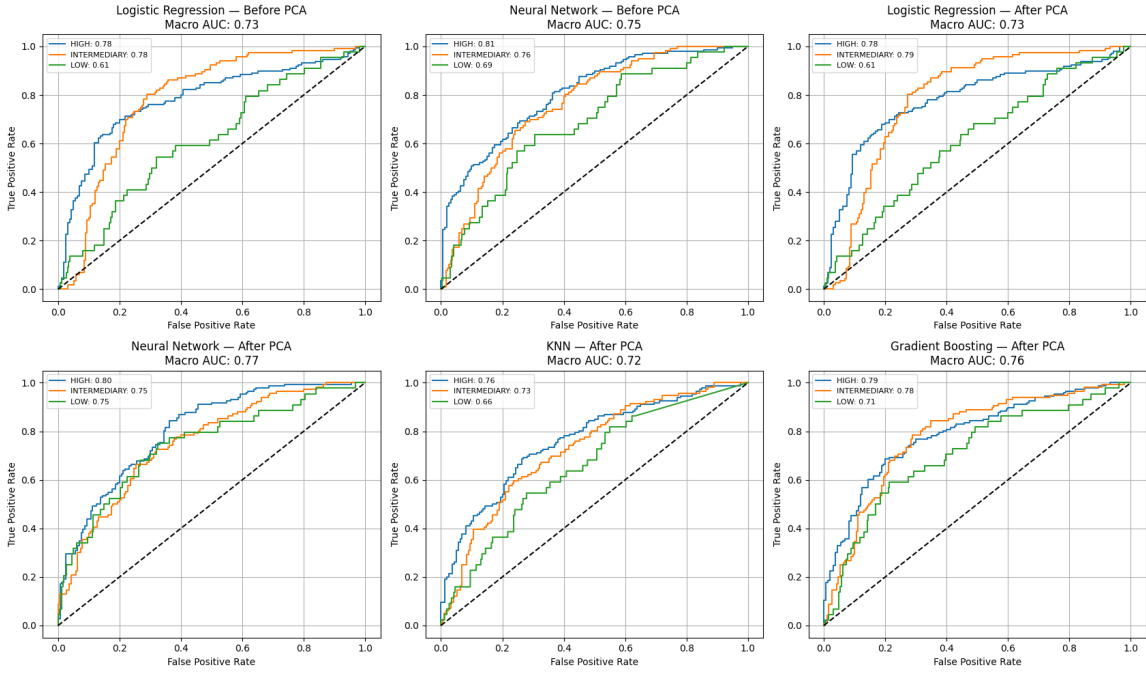


Fig. 2. One-vs-rest ROC curves and per-class AUC for six model/PCA combinations.

TABLE VIII
REGRESSION PERFORMANCE FOR CVD RISK SCORE

Metric	Value
MAE	0.408
RMSE	0.544
R^2	0.951

threshold tuning as a low-cost way to recover LOW-class recall.

E. CVD Risk Score Regression

Table VIII reports the Random Forest Regressor’s performance predicting the continuous CVD Risk Score: $R^2 = 0.951$ with a mean absolute error of only 0.408 on a score ranging roughly 10–24. Fig. 3 plots predicted against actual scores, showing points tightly clustered around the ideal diagonal with only minor deviation at the extremes; this is consistent with the $R^2 \approx 0.96$ reported by Tasnim et al. on the same 1,529-patient dataset [12], confirming that the continuous score is substantially easier to predict than its three-way discretization. Residuals were centered near zero and mostly within ± 1 point across the prediction range, with no systematic bias, though a small number of larger residuals (up to ± 2 points) clustered around the middle of the range (14–20) where most data is concentrated—consistent with the LOW/INTERMEDIARY/HIGH boundary difficulty discussed in Section VII-A.

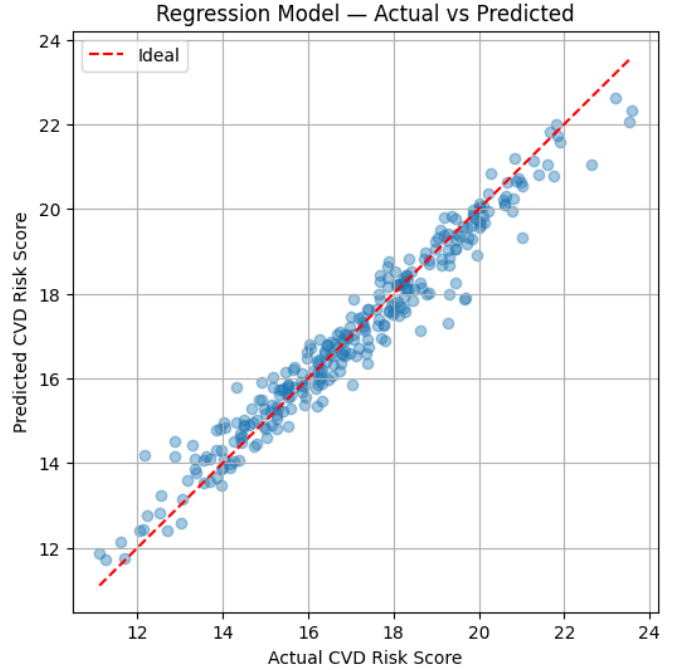


Fig. 3. Regression model—actual vs. predicted CVD Risk Score on the test set.

VII. DISCUSSION

A. The LOW-Risk Class Remains the Main Bottleneck

Across every classifier, the LOW-risk class was hardest to identify, and two of the four baseline models never predicted it at all. This is only partly explained by its smaller sample size

(14.4% of records): the much stronger regression performance (Table VIII) shows the underlying CVD Risk Score is well predicted throughout its range, suggesting LOW-risk patients are not inherently harder to characterize but sit close to the INTERMEDIARY boundary once the continuous score is thresholded into three classes—consistent with the LOW-INTERMEDIARY confusion in Fig. 1.

B. Why the Simplest Model Performed Best

Logistic Regression, the simplest model evaluated, achieved the highest accuracy, narrowly ahead of Gradient Boosting and AdaBoost. Given that the CVD Risk Score is well explained by an approximately linear combination of clinical predictors ($R^2 = 0.951$), a linear decision boundary transfers reasonably well to the discretized target. The from-scratch Random Forest was also intentionally kept shallow to remain tractable, limiting its capacity relative to a more heavily tuned ensemble. Notably, Logistic Regression’s ranking ability (macro AUC 0.73) is not the best among the six model/PCA combinations—its advantage lies in how well its default 0.5 threshold aligns with the HIGH/INTERMEDIARY boundary, not in superior overall class separation.

C. Overfitting in the Neural Network

The Neural Network showed a clear overfitting pattern, with training accuracy climbing toward 0.80 while validation accuracy plateaued within the first 15–20 epochs. Early stopping together with dropout or L2 regularization would likely narrow this gap. Despite the overfitting, the network still attained the best macro AUC after PCA (0.77) and the best LOW-class recall among the baseline models, suggesting its probability estimates remain informative even where raw accuracy trails Logistic Regression.

D. Limited Benefit of Dimensionality Reduction

PCA’s mostly neutral-to-slightly-negative effect on accuracy (Table V) indicates that, aside from a few clearly redundant pairs (e.g., height in meters versus centimeters, cholesterol versus LDL), the 27-dimensional feature space does not contain enough linear redundancy for a 16-component projection to meaningfully simplify the problem. PCA does not qualitatively change which classes are confused—LOW is still absorbed mainly into INTERMEDIARY—so further gains are more likely to come from better modeling of existing features than from further dimensionality reduction.

E. Comparison with Related Work

Our best classification accuracy of 65.36% is lower than the 68.22% XGBoost accuracy reported by Islam et al. on a related Bangladeshi dataset [10], and considerably lower than the C-statistics reported for binary or survival-based risk models in Section II [1], [2], [4]. This gap is expected: three-way multiclass risk stratification is a strictly harder problem than binary classification, and our project intentionally implements Random Forest and AdaBoost from scratch with modest ensemble sizes for pedagogical transparency rather

than maximum performance. The strong regression R^2 of 0.951, close to the 0.96 reported by Tasnim et al. using the same dataset [12], indicates that the gap is concentrated in the class-boundary regions introduced by discretizing the score rather than in the underlying relationship between the clinical features and cardiovascular risk.

VIII. CONCLUSION AND FUTURE WORK

A. Conclusion

This project compared machine learning approaches for cardiovascular risk stratification on a dataset of 1,529 patients described by 18 clinical, demographic, and lifestyle features. Among four baseline classifiers, Logistic Regression achieved the highest test accuracy (65.36%), narrowly ahead of AdaBoost (64.38%) and the Neural Network (62.42%), with the from-scratch Random Forest trailing at 58.82%. Extending the comparison with Gradient Boosting on PCA-reduced features (65.03%) and KNN (59.80%) did not change the overall ranking meaningfully. PCA had a small and inconsistent effect on accuracy, and the LOW-risk class remained the hardest to identify across every model, with Random Forest and AdaBoost failing to recall it at all, as confirmed by the confusion matrices in Fig. 1. ROC analysis showed that several models, particularly the Neural Network and Gradient Boosting, still separate the LOW class reasonably well by probability (AUC up to 0.75), indicating the recall problem is partly a threshold issue rather than a purely representational one. In contrast, a Random Forest Regressor trained on the same features to predict the continuous CVD Risk Score achieved an R^2 of 0.951 with residuals tightly and evenly distributed around zero, showing that the underlying clinical relationship is strongly learnable and that most of the classification difficulty stems from discretizing that score into three risk levels rather than from the predictors themselves.

B. Future Work

Future work could address the class imbalance more directly through oversampling techniques such as SMOTE, class-weighted loss functions, or decision-threshold tuning informed by the per-class ROC curves in Section VI-D, particularly for the LOW-risk group. Replacing the from-scratch Random Forest and AdaBoost with more heavily tuned ensembles, or introducing gradient-boosted methods such as XGBoost or LightGBM, may close some of the remaining accuracy gap. Given the strong regression performance observed in Section VI-E, a hybrid strategy that first predicts the continuous CVD Risk Score and then derives the discrete risk level from tuned thresholds could outperform direct multiclass classification, and treating the three risk levels as ordinal rather than purely categorical may also better reflect the LOW/INTERMEDIARY boundary effect. Finally, adding early stopping and regularization to the Neural Network, and validating all models on an external cohort, would help confirm that these results generalize beyond the current dataset.

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